ASSUMPTIONS & CONVENTIONS

STATUS — olympiad/specialised extension (IOQM/RMO-flavoured)

The official JEE Advanced syllabus specifies the binomial theorem for **positive integral index only**. The general series below is extension material (it underlies JEE Main-style approximation questions historically, and all of analysis).

KEY RESULT

The generalised coefficient and Euler's series

For any real r, $\binom{r}{k} = \frac{r(r-1)\cdots(r-k+1)}{k!}$ (with $\binom{r}{0} = 1$), and

$$(1+x)^r = \sum_{k=0}^{\infty} \binom{r}{k} x^k \qquad \text{for } |x| < 1$$

The extension is forced: it is the unique choice keeping $\frac{d}{dx}(1+x)^r = r(1+x)^{r-1}$ term-by-term. Families to own: $(1-x)^{-n} = \sum_k \binom{n+k-1}{k} x^k$ (stars-and-bars counts — coefficients of $(1-x)^{-3}$ are $\binom{k+2}{2}$: x^2 coefficient 6); $(1-x)^{-1/2} = \sum_k \frac{\binom{2k}{k}}{4^k} x^k$ (x^5: $\frac{252}{1024} = \frac{63}{256}$);
$(1+x)^{1/2} = 1 + \frac{x}{2} - \frac{x^2}{8} + \frac{x^3}{16} - \cdots$

KEY RESULT

Validity is part of the answer

Factor so the bracket is $1+u$ with $|u| < 1$.
$(3+2x)^{-2} = \frac{1}{9}\left(1+\frac{2x}{3}\right)^{-2} = \frac{1}{9} - \frac{4x}{27} + \cdots$ valid for $|x| < \frac{3}{2}$; in descending powers factor $2x$ instead, valid for $|x| > \frac{3}{2}$.
$(4+3x)^{-1/2} = \frac{1}{2} - \frac{3x}{16} + \cdots$ for $|x| < \frac{4}{3}$.
Coefficient of x^6 in $(1+3x)^{-2}$: $(-1)^6 \cdot 7 \cdot 3^6 = 5103$, valid $|x| < \frac{1}{3}$.

WORKED EXAMPLE

Worked — approximation with an error budget

$\sqrt[3]{26} = 3\left(1-\frac{1}{27}\right)^{1/3} \approx 3\left(1-\frac{1}{81}-\frac{1}{6561}\right) = 2.9$; the first dropped term is $< 10^{-4}$ in the final value, certifying 2.96.
$\sqrt{1.02} \approx 1 + 0.01 - 0.00005 + 0.0000005$: with first dropped term $< 10^{-8}$. Series recognition in reverse: $\sum_k (k+1)(0.9)^k = \frac{1}{(1-0.9)^2} = 100$, and $\sum_k \frac{\binom{2k}{k}}{8^k} = \left(1-\frac{4}{8}\right)^{-1/2} = \sqrt{2}$.

COMMON TRAP

Common traps — general series

- Writing an expansion without its validity range: zero marks, even with correct coefficients.
- The radius is the ratio inside the bracket, never automatically 1.
- Boundary $x = \pm 1$ is delicate (convergence depends on r); JEE-level questions test $|x| < 1$ only — do not evaluate at endpoints without Abel-type justification.

ASSUMPTIONS & CONVENTIONS

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Modular properties of binomial coefficients (Lucas, Kummer, prime rows) are Olympiad number theory built on the Part I census.

KEY RESULT

Prime rows and the Freshman's dream

For prime p and $0 < k < p$: $p \mid \binom{p}{k}$ (the numerator carries p, the denominator cannot cancel it). Hence in characteristic p,

$$(a+b)^p \equiv a^p + b^p \ (\text{mod } p)$$

and iterating, $(a+b)^{p^m} \equiv a^{p^m} + b^{p^m} \ (\text{mod } p)$. Row p of Pascal's triangle mod p is $1, 0, 0, \ldots, 0, 1$ — brute-force verified for all primes ≤ 47 in math_checks.

KEY RESULT

Lucas's theorem (statement and use)

Write n, k in base p: $n = \sum n_i p^i$, $k = \sum k_i p^i$. Then $\binom{n}{k} \equiv \prod_i \binom{n_i}{k_i}$ (extmodp) (with $\binom{n_i}{k_i} = 0$ if $k_i > n_i$). Consequence: $\binom{n}{k}$ is **odd** iff every binary digit of k is $\leq$ that of n (k is a submask of n) — so row n contains exactly $2^{(\text{number of 1 bits in } n)}$ odd entries. Kummer (statement): the exponent of p in $\binom{n}{k}$ equals the number of carries when adding k and $n-k$ in base p.

KEY RESULT

Finite differences annihilate polynomials

The n-th forward difference of a degree-n polynomial is the constant $n! \times$ (leading coefficient):

$$\sum_{k=0}^{n} (-1)^{n-k} \binom{n}{k} P(x+k) = n! \times [\text{leading coe}$$

. For $P(x) = x^n$:

$$\sum_{k} (-1)^k \binom{n}{k} (x+k)^n = (-1)^n n!$$, independent of x — a pure binomial-theorem telescoping, checked numerically in math_checks.

COMMON TRAP

Common traps — modular binomials

- $(a+b)^p \equiv a^p + b^p$ is modulo p (or in characteristic p) — never an integer equality.
- Lucas needs base-p digits of **both** n and k aligned; a single digit $k_i > n_i$ kills the coefficient mod p.
- $\binom{p}{k}$ divisible by p requires $0 < k < p$ strictly; endpoints are 1.